

Dr. Meena Rao

Ph.D., M.Tech., B.E. | Associate Professor

Department of Electronics & Communication Engineering, Maharaja Surajmal Institute of Technology (MSIT), New Delhi

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PROFESSIONAL SUMMARY

Accomplished Academician and Researcher with over 21 years of experience in higher education, advanced research, and institutional administration. Proven expertise in Artificial Intelligence, Machine Learning, IoT, and MANETs with a prolific publication record in SCI/Scopus-indexed journals and granted patents. Actively serves as Co-Convener and former Convener (Sep 2017 – June 2026) of the Placement Committee, driving campus recruitment drives and robust corporate HR relations.

AREAS OF EXPERTISE

Artificial Intelligence	Machine Learning	Internet of Things (IoT)	MANETs Electronics	Digital
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WORK EXPERIENCE

Associate Professor | Department of Electronics & Communication Engineering June 2017 – Present
Maharaja Surajmal Institute of Technology (MSIT), New Delhi

- Deliver core and advanced ECE courses, guide student research projects, and drive departmental academic strategy.
- Serves as Co-Convener (present) and former Convener (Sep 2017 – June 2026) of the Placement Committee, spearheading campus placement drives for B.Tech students and maintaining long-term corporate linkages with HR leaders.

Assistant Professor | Department of Electronics & Communication Engineering Jan 2005 – June 2017
Maharaja Surajmal Institute of Technology (Affiliated to GGSIPU), New Delhi

- Conducted undergraduate lectures, laboratory sessions, and curriculum development across key ECE disciplines for over a decade.

EDUCATION & PROFESSIONAL TRAINING

Educational Qualifications	Certifications & Training
Ph.D. in Electronics & Communication Engg. (2016) Gautam Buddha University Thesis: Routing Protocols for QoS Provisioning in MANETs	Certificate of Merit (AI): IIT Kharagpur AI4ICPS & TCS iON HAAI Programme.
M.Tech. in Digital Electronics (2005) Visvesvaraya Technological University (VTU), Karnataka	NPTEL Certifications: Industry 4.0 & Industrial IoT; Internet of Things.
B.E. in Electronics & Communication Engg. (2003) Visvesvaraya Technological University (VTU), Karnataka	ML Internship: NITK Surathkal & Pantech E-Learning. Deep Learning: Two-Week Industrial Training Programme.

PATENTS & AUTHORED BOOKS

- Published Patents:** GPS Data Spoofing and Malfunctioning Detection System Using Classifiers (Patent Application No: 2021100964; AI-enabled Cyber Command Console (Issue Date: 2026); AI, Big Data, IoT & Blockchain in Smart Energy Management (Pub Date: 2024); Intelligent Vehicle-Controlled Rescue System (2021) etc.
- Books Authored:** Authored 3 books covering MANET QoS Routing Protocols, FTP Access Management, and Automobile Recognition Devices for Blindness.
- Book Chapters:** Contributed peer-reviewed book chapters for international publisher collections (Springer / CRC Press series) covering advanced Machine Learning architectures, IoT network security, and intelligent routing protocols.

SOME KEY RESEARCH PUBLICATIONS

- **Meena Rao**, Parul Chaudhary, Kavita Sheoran, Geetika Dhand, “A secure routing protocol using hybrid deep regression based trust evaluation and clustering for mobile adhoc network”, Peer-to-Peer Networking and Applications, Springer, Vol. 16, pp. 2794–2810, Oct 2, 2023. [SCIE, Scopus, UGC Indexed]
- Geetika Dhand, **Meena Rao**, Parul Chaudhary and Kavita Sheoran, “A Secure Routing and Malicious Node Detection in Mobile Ad Hoc Network Using Trust Value Evaluation with Improved XG Boost Mechanism”, Journal of Computer and Network Applications, Elsevier ScienceDirect, Dec 10, 2024. [SCIE, ISSN: 1084-8045]
- **Meena Rao** and Neeta Singh, “Energy Efficient QoS Aware Hierarchical KF-MAC Routing Protocol in Manet”, Wireless Personal Communications, Springer Link, Vol. 80, No. 3, pp. 1-14, Apr 12, 2018. [SCIE, Scopus, UGC-CARE Indexed, ISSN: 0929-6212]
- Richa Gupta, Deepshikha Yadav, **Meena Rao** and Ankit Gaur, “Design and Testing of SMPS in Accordance with IEC 61000-4-11 Standard with Measurement Uncertainty”, MAPAN, Springer, Vol. 37, pp. 71-80, Aug 27, 2021. [SCIE, Scopus, UGC-CARE Indexed]
- Sameer Dixit, Priyanka Pande, Shubham Sahai, **Meena Rao**, Jitendra Kumar, “Adaptive Intrusion Detection System for Wireless Sensor Networks Using Deep Learning Models”, Panamerican Mathematical Journal, Vol. 35, No. 4s, 2025. [Scopus Indexed, ISSN: 1064-9735]
- Parul Chaudhary, **Meena Rao** and Parveen Kumar, “Design of Digital Differentiator Using Teacher Learning Based Optimization Algorithm”, Presented in National Conference for Integrating Technologies, Ideas and Disciplines for Engineering Innovation (NCI-TIDE 2025), Nov 18-19, 2025. [Awarded Best Paper]
- Parul Chaudhary, **Meena Rao**, “Intelligent Aid: A tool for better communication and navigation for the visually challenged”, Presented in 1st IEEE International Conference on Advances in Computing, Communication and Networking (ICAC2N-2024), Dec 16-17, 2024.
- Richa Gupta, Deepshikha Yadav, **Meena Rao**, Ankit Gaur, “Design & testing of SMPS in accordance with IEC-61000-4-11 standard with lower measurement uncertainty”, Presented in 7th National Conference on Advances in Metrology (AdMet 2021), Springer, MSIT, New Delhi, March 5-6, 2021.
- **Meena Rao**, “Impact of Routing Protocols and Artificial Intelligence Techniques on adhoc networks in critical scenario”, 2nd International Conference on Emerging Technologies in Data Mining and Information Security, Lecture Notes in Networks and Systems, Springer, Vol. 164, pp. 883-890, July 2-4, 2020

AWARDS & PROFESSIONAL SERVICE

- **Best Paper AwardS:** Awarded for research paper at NCI-TIDE 2025 Conference and 2nd Int. Conference on Emerging Tech in Data Mining (Springer, 2020).
- **Institutional Excellence:** Awarded Certificate of Excellence in Research (MSIT, 2022 & 2023) and Letters of Appreciation (2016, 2024).
- **Academic Leadership:** Served as Session Chair at NCI-TIDE 2025 and 10th Int. Conference on Microelectronics, Computing & Communication.
- **Professional Memberships:** Life Member - Indian Society for Technical Education (ISTE) | Distinguished Fellow Member - Indian Society for VLSI Education (ISVE).